Q1: Script to Find World-Writable Files

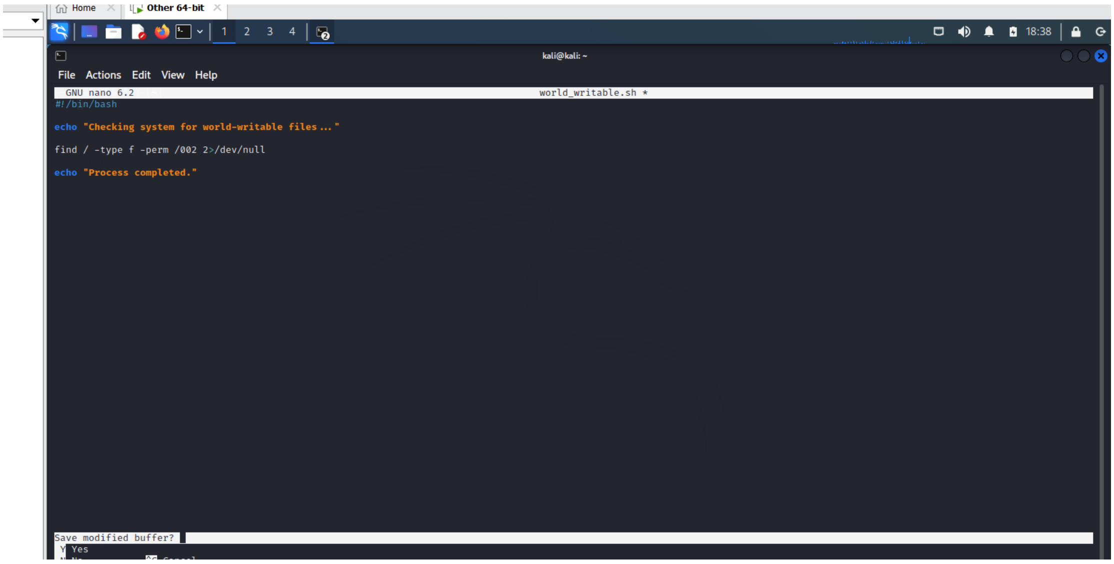
Script:

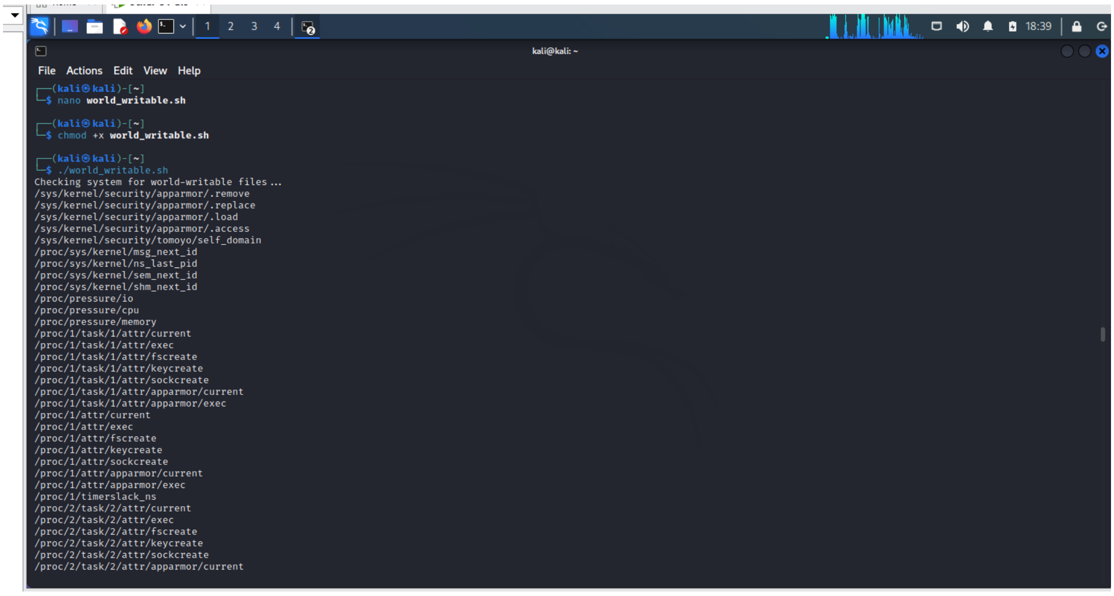
#!/bin/bash

echo "Checking system for world-writable files..."

find / -type f -perm /002 2>/dev/null

echo "Process completed."





Explanation

In this question, I wrote a bash script to locate files that have write permission enabled for all users. These types of files are called world-writable and are considered unsafe in most cases.

The script uses the `find` command to search the entire system. The option `-type f` ensures that only regular files are included. The permission `/002` is used to match any file where the write permission is given to others.

Since some directories are restricted, I used `2>/dev/null` to ignore error messages and display only valid results.

This script is useful in security checking because it helps identify files that could be modified by unauthorized users.

Q2: How Windows NTFS Implements DAC

Answer:

Windows NTFS follows the Discretionary Access Control (DAC) model to manage permissions for files and directories. In this model, access control decisions are handled by the file owner.

Every file in NTFS contains a set of security details, including an Access Control List (ACL). This list is made up of multiple entries that define permissions for different users and groups.

Each entry specifies what actions are allowed, such as opening, editing, or executing a file. When someone tries to access a file, the system reads these entries and decides whether the request should be allowed or denied.

The key feature of DAC is flexibility, as the owner can update permissions at any time. However, this flexibility can also lead to security risks if permissions are not set carefully.